YEAR 11 MATHS METHODS INVESTIGATIVE TASK 1

Functions

Prerequisites
Knowledge of:

- Linear, quadratic (parabolic), polynomial, rectangular, hyperbolic and circle graphs
- The domain and range of a graph
- Gradient, midpoint and length of a straight line segment
- Curve fitting.

[Total: 90 marks]

1 For each of the following graphs:

i Describe the general shape of the graph.

ii List the main features of the graph you should find (e.g. gradient, intercepts, turning point etc.) You may use your notes.

iii Sketch the graph, showing and/or describing these features. CAS calculators should not be used for this question. Graph paper may be used, but is not essential as long as the main features are clear.

a $y = 2x - 6$

b $y = x^2 + 2x - 8$

c $(x + 1)^2 + (y - 3)^2 = 16$

d $xy + 2 = 0$

e $y = (x - 2)^2(x + 1)(x + 3)$

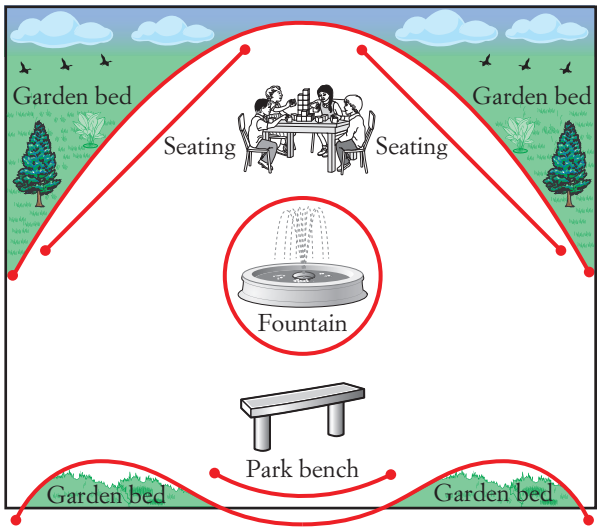
[8 + 13 + 6 + 9 + 8 = 44 marks]

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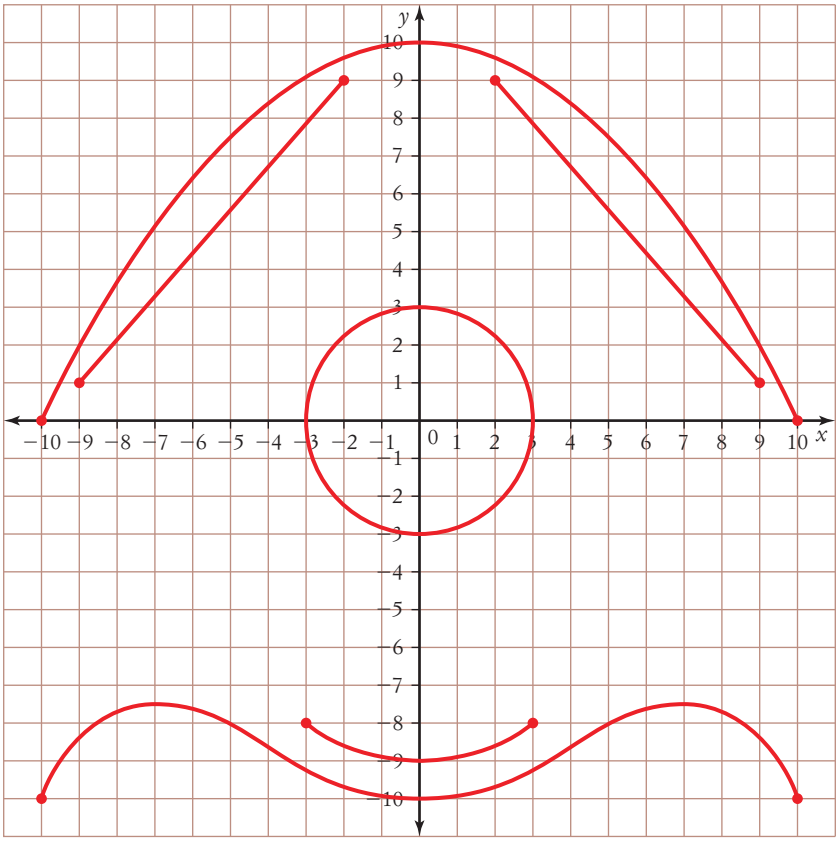
The Garden

Questions **2** to **5** apply the garden scenario which follows.

A landscaper has designed a square garden for a Mathematician, Mr Bean. Each side is 20 m in length. The landscaper has lost the original equations given to him but has his working sketch. The landscaper was given a little leeway, so any equation that is very similar to the design will work.



The landscaper was good at Maths in school and remembered that the equation of the two southern gardens was a quartic of the form $y = -kx^2(x - a)(x - b) - c$. He also remembered that the border of the northern gardens was a parabola. Use the garden plan below to answer the following questions.



2 The mathematician wanted to place six statues, one at each end and one in the middle of both sets of seating. What is the position of each statue? **[3 marks]**

3 If the seating uses long lengths of strong timber, how long, to the nearest cm, should each piece of timber be? **[3 marks]**

4 The Mathematician's daughter Amy wanted her own small circular garden bed with an area of at least 10 square metres. The landscape gardener decided on a radius of 2 m. Find a suitable place for Amy's garden bed. Her father decided the garden would be too cluttered if the bed were too close to the y-axis.

a Is a radius of 2 m sufficient to give the required area of at least 10 m^2?

b Find the equation of a suitable plot for Amy's garden bed.

[1 + 3 = 4 marks]

5 Use the garden plan in the diagram to find the equations of the position and shape of the following. Include the domain for restricted graphs.

a The seating

b The circular fountain

c The curved (parabolic) section of the two sets of garden beds on the north

d The circular park bench, giving both domain and range.

e The border of the southern gardens

[8 + 3 + 5 + 10 + 10 = 36]

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